PEPII

RF SYSTEM

LLRF FEEDBACK LOOP DESCRIPTION

PS-340-330-52-R0

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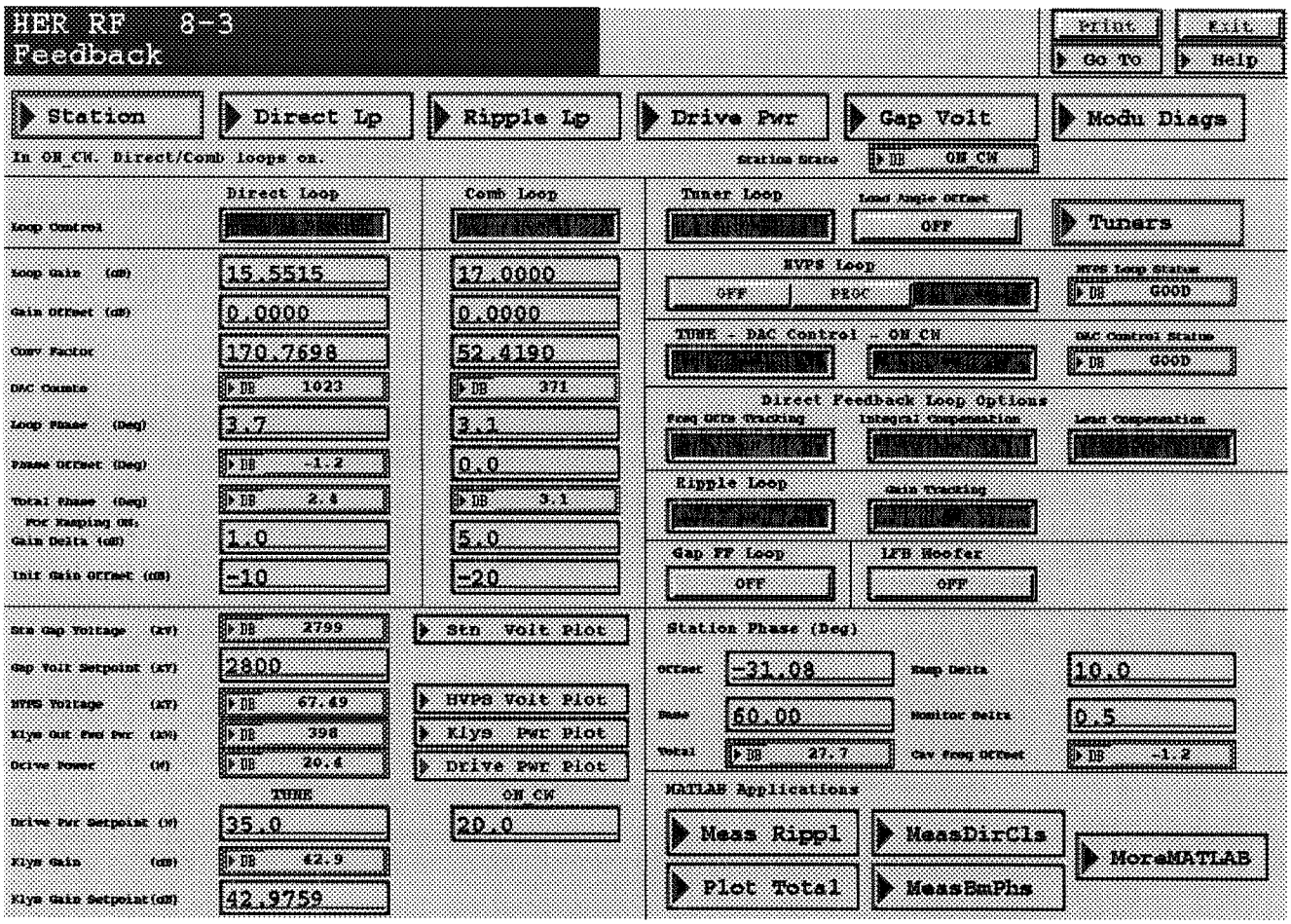
PEP-II LLRF FEEDBACK LOOP DESCRIPTION

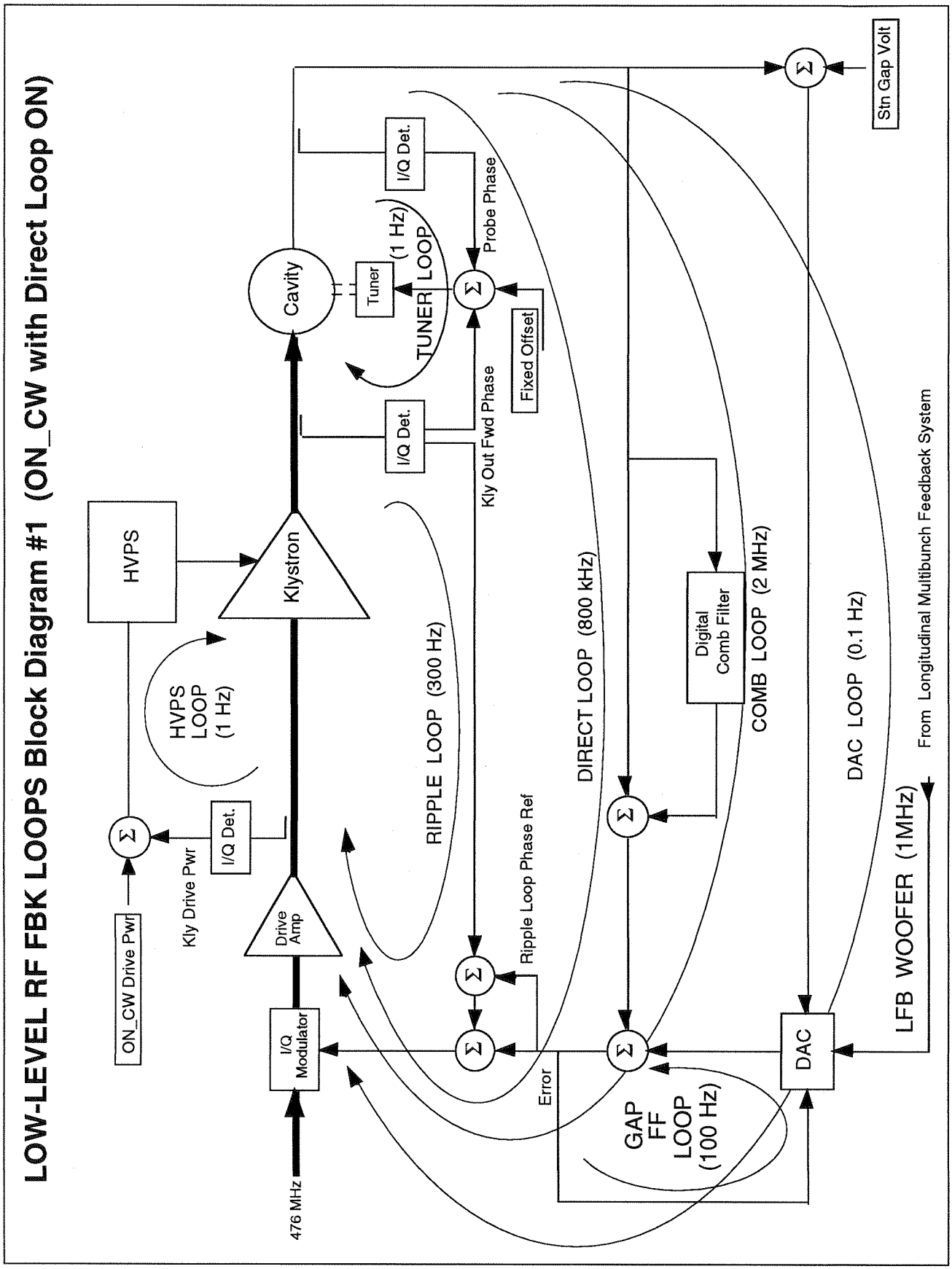
1) List of Loops:

DIRECT LOOP
COMB LOOP
TUNER LOOP
HVPS LOOP
DAC LOOP
DIRECT FEEDBACK LOOP OPTIONS
RIPPLE LOOP
GAP FF LOOP
LFB WOOFER
OPTIMIZED STATION PHASING ROUTINE

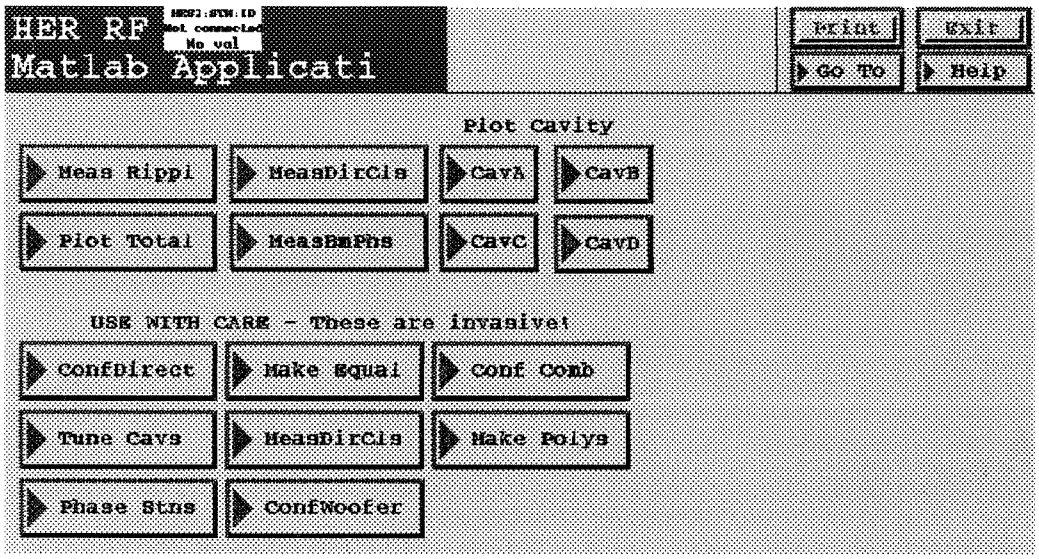
2) NOTES:

Note: All loops are described in the order their buttons appear on the **EPICS Feedback** panel.
See also Block Diagram: **LOW-LEVEL RF FBK LOOPS Block Diagram #1.**




LOW-LEVEL RF FBK LOOPS Block Diagram #1 (ON_CW with Direct Loop ON)
HVPS
Klystron
Cavity
Tuner
(1 Hz)
TUNER LOOP
I/Q Det.
I/Q Det.
I/Q Det.
Probe Phase
Fixed Offset
Kly Out Fwd Phase
Stn Gap Volt
HVPS LOOP (1 Hz)
RIPPLE LOOP (300 Hz)
DIRECT LOOP (800 kHz)
Digital Comb Filter
COMB LOOP (2 MHz)
DAC LOOP (0.1 Hz)
From Longitudinal Multibunch Feedback System
LFB WOOFER (1MHz)
ON_CW Drive Pwr
Kly Drive Pwr
Drive Amp
I/Q Modulator
476 MHz
Ripple Loop Phase Ref
Error
GAP FF LOOP (100 Hz)
DAC

Note: All MATLAB routines are accessible through the Matlab Applications panel. For all MATLAB calibration routines the station must be in **ON_CW**.



3) Normal Operation for Stored Beam (DIRECT LOOP ON):

DIRECT LOOP

The DIRECT LOOP is required for lowering the cavity impedance to reduce multi-bunch oscillations of the beam. Functionally the direct loop keeps the gap voltage constant as set by a DAC Reference over an 800 kHz bandwidth. The loop compares the combined baseband field signals of a station's cavities to the reference generated by the Gap module. The resultant error signal is up-converted to RF and drives the klystron.

The DIRECT LOOP contains a PID controller with an INTEGRAL COMPENSATION for smoothing out the ripple caused by the klystron high voltage power supply and LEAD COMPENSATION that increases the bandwidth and gain of the loop.

The FREQUENCY OFFSET TRACKING loop takes out the phase shift caused by detuning of the cavities during heavy beam loading. It is used as a diagnostic for adjusting the waveguide network and should not normally be activated.

A MATLAB calibration routine sets up the DIRECT LOOP for proper loop phase, loop gain and gain tracking initiated by the *"ConfDirect"* button.

The closed loop response of the DIRECT LOOP (and COMB LOOP) can be measured using the built in network analyzer by initiating the MATLAB routine *"MeasDirCls"*. This function will not cause loss of stored beam.

COMB LOOP

The COMB LOOP provides additional impedance reduction for the cavities at specific synchrotron frequency sidebands around the revolution harmonics of the beam. It operates over a bandwidth of 2 MHz and includes a 1 turn delay.

A MATLAB calibration routine sets up the Comb Loop for proper loop phase, gain and delay, initiated by the *"Config Comb"* button. The station must be in ON_CW with the DIRECT LOOP closed.

Another MATLAB routine makes an equalizer to compensate for the effects of group delay, initiated by the *"Make Equal"* button. The equalizer is used both in the COMB LOOP and in the WOOFER link. The user should never have to create new equalizer files.

The closed loop response of the DIRECT LOOP and COMB LOOP can be measured using the built in network analyzer by initiating the MATLAB routine *"MeasDirCls"*. This function will not cause loss of stored beam.

TUNER LOOP

The TUNER LOOP tunes and maintains each cavity at resonance. It corrects for thermal frequency variations and compensates cavity beam loading by keeping the phase relationship between forward power and cavity field, as seen by the cavity probe, constant. The relevant phases are measured by digital IQ detectors and the loop is completed in software controlling the tuner position via a stepping motor.
To establish the resonance condition for each cavity a MATLAB routine is used which is initiated by the *"Tune Cavs"* button. In several iterations the routine measures the resonance curves of all cavities of the station by injecting noise onto a CW RF signal and then fitting standard resonance curves to the measured resonance response. From these fitted resonance curves the program then establishes resonance. The program also establishes the correct vector summation of the multiple cavity signals at the baseband level for use in the Direct and Comb Loops.
A similar MATLAB routine (*"Make Poly"*) is used to make a polynomial of the resonance frequency versus tuner position, information needed to park the cavities at the desired park frequency and to allow the direct loop to adjust phase as the cavities detune.

HVPS LOOP (DIRECT LOOP ON)

The HVPS LOOP adjusts the voltage to the klystron to provide sufficient output power to operate the station under whatever gap voltage or beam loading is requested. Functionally the loop keeps the klystron operating at about 10% below saturated output power. The loop measures the drive power at the input to the klystron and compares it to the ON_CW Drive Power set-point. Based on the error the set-point for the high voltage power supply is adjusted up for excessive drive and down for insufficient drive. This is a slow loop with about a 1 Hz bandwidth.

DAC LOOP

The DAC Loop is a slow (0.1 Hz bandwidth) loop in software which functionally keeps the measured gap voltage of the station equal to it's requested "Station Gap Voltage" by adjusting the DAC in the Gap Voltage Feed-Forward module.

DIRECT FEEDBACK LOOP OPTIONS

The DIRECT FEEDBACK LOOP OPTIONS control the optional functions of the DIRECT LOOP: FREQUENCY OFFSET TRACKING, INTEGRAL COMPENSATION and LEAD COMPENSATION (see DIRECT LOOP description).

RIPPLE LOOP

The RIPPLE LOOP is intended to remove amplitude and phase ripple in the klystron output power but at the time it is only utilized to keep the low bandwidth phase across the klystron and drive amplifier constant as the klystron voltage is varied. The RIPPLE LOOP should be ON for all normal operation.

GAP FF LOOP

The GAP FEED FORWARD LOOP is required to tell the DIRECT LOOP to ignore the effects of the ion-clearing gap in the beam bunch train. Functionally the loop learns about the variation in the klystron drive caused by the beam gap and adds an equal variation in the reference signal so that the error signal driving the klystron stays unchanged. This loop adapts fully in about 1000 beam revolutions.

LFB WOOFER

The LONGITUDINAL FEEDBACK WOOFER is the third cavity impedance reduction loop along with the DIRECT LOOP and the COMB LOOP. It derives it's information from the lowest beam oscillation modes detected by the Longitudinal Multibunch Feedback system and uses one RF station in each ring as a powerful longitudinal kicker.
A MATLAB calibration routine sets up the Woofer Loop for proper loop phase, loop gain and one turn delay, initiated by the *"ConfWoofer"* button. The station must be in ON_CW with the DIRECT LOOP, COMB LOOP and GAP FF LOOPS operating.

OPTIMIZED STATION PHASING ROUTINE

A MATLAB optimization routine can be utilized to set the correct phase for each station for maximum voltage gain, initiated by the *"Phase Stns"* button. This routine is operational only if the beam current is above 100 mA. It equalizes the power contribution of each operational station in a ring by changing the station phase in 1/2 degree maximum steps for 10 iterations. One station in each ring is designated as reference station and its phase stays fixed. If optimum phasing is not achieved in one try the routine can be repeated. In the HER station 8-3 is used as reference with station 12-3 as alternate if 8-3 is off. In the LER station 4-4 is the reference station.

4) Alternate Station Operating Conditions:

The following loops are configured differently when the Direct Loop is OFF (the station is in ON_CW or TUNE or ON_FM). These operating modes are usually used when there is no beam stored.

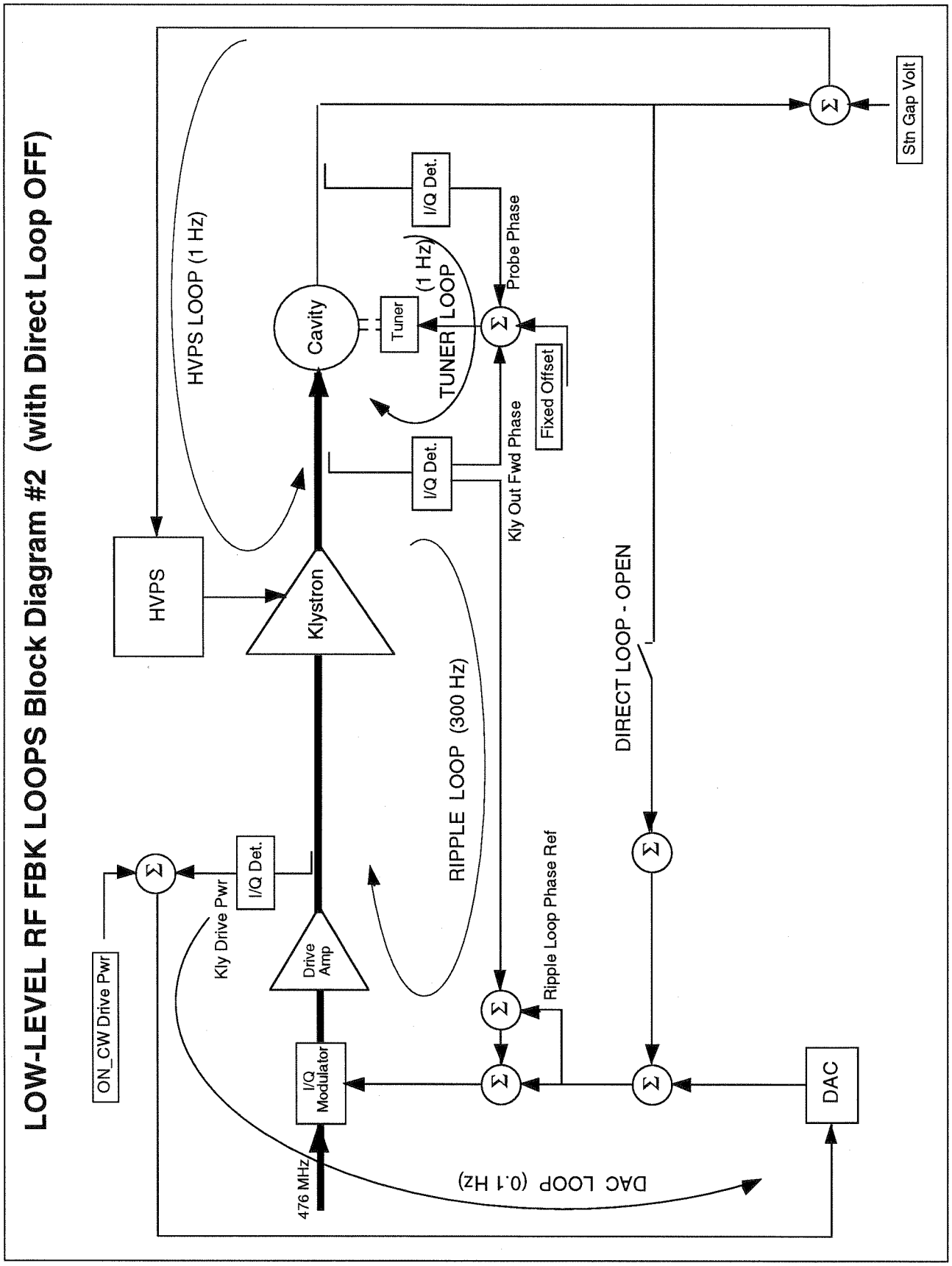
See Block Diagram: **LOW-LEVEL RF FBK LOOPS Block Diagram #2.**

HVPS LOOP

The HVPS LOOP in the Direct Loop OFF mode functionally keeps the measured gap voltage of the station equal to it's requested "Station Gap Voltage" by adjusting the high voltage of the klystron and as a result the klystron output power.

DAC LOOP

The DAC LOOP in the Direct Loop OFF mode functionally keeps the drive power at the requested drive level by adjusting the DAC in the Gap Voltage FF module.


LOW-LEVEL RF FBK LOOPS Block Diagram #2 (with Direct Loop OFF)
HVPS LOOP (1 Hz)
HVPS
Klystron
Cavity
Tuner
TUNER LOOP (1 Hz)
I/Q Det.
I/Q Det.
Probe Phase
Fixed Offset
Kly Out Fwd Phase
Stn Gap Volt
DIRECT LOOP - OPEN
RIPPLE LOOP (300 Hz)
Ripple Loop Phase Ref
ON_CW Drive Pwr
Kly Drive Pwr
I/Q Det.
Drive Amp
I/Q Modulator
476 MHz
DAC
DAC LOOP (0.1 Hz)